

Harsh Bafna

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PROFESSIONAL SUMMARY

Machine Learning Engineer with ~3 years of experience specializing in Generative AI, multi-agent system orchestration, and production-grade LLM applications. Proven track record of architecting scalable retrieval (RAG) systems and leading engineering teams to deploy high-impact AI solutions.

EXPERIENCE

Machine Learning Engineer, Quantiphi | Bengaluru, IN Feb 2024 – Present

RCA, Segmentation & NL2SQL Bot Python, SQL, ADK, LLMs, AI Agents, Multi-Agent Orchestration

- Led a team of 5+ ML engineers within a 24+ member team to design and ship 3 production-grade multi-agent orchestration systems, enabling non-technical users and C-suite executives to query enterprise data through natural-language chat.
- Architected and deployed a multi-agent analytics platform on Google’s ADK framework with a custom RAG-based retrieval layer over table schemas, spanning 14+ structured data sources (Kafka topics, Google Ads, Google Analytics) to power accurate, context-aware NL2SQL generation.
- Cut executive data-reporting and analysis turnaround from days to minutes, accelerating data-driven decision-making across the organization.

AI Search & Recommendation Engine Python, Vertex AI Search, LLMs, Ranking

- Designed and architected a conversational search and recommendation engine, replacing traditional keyword-based search with natural-language query understanding.
- Engineered a conversational search platform leveraging Vertex AI Search for Commerce, indexing an 80K+ product SKU catalog, alongside recommendation models (recommended-for-you, similar items, frequently-bought-together) to surface more relevant results.
- Elevated search from keyword/fuzzy matching to free-form conversational queries, improving product-catalog sorting accuracy against user intent.

AI Email Campaign Generator Python, LLMs, AI Workflow, Web-scraping, NLP, Imagen

- Architected an AI-driven email campaign generator that scrapes theme and metadata from property websites and applies it to auto-generated marketing emails.
- Built scheduled extraction jobs to pull property metadata/themes and an AI workflow to generate each section of the email, including image enhancement via Imagen models.
- Cut email-campaign drafting time for analysts from days/weeks to minutes by automating theme extraction and end-to-end content generation.

Lead Hotspot Detection Python, LangChain, LLMs

- Developed and evaluated multiple ML models to predict lead-contamination hotspots, selecting the best-performing hybrid approach.
- Cleaned and engineered features from PWS data sourced from the SDWIS government database, performing EDA and combining model-driven predictions with rule-based decision logic.
- Achieved 88+% accuracy in lead hotspot detection, driving more effective resource allocation for remediation efforts.

Software Engineer Intern, JP Morgan Chase & Co. | Remote Nov 2022 – Sept 2023

- Designed a real-time data streaming interface capable of ingesting and processing over 100,000 data points per second.
- Utilized JP Morgan Chase’s Perspective open-source data visualization tool to build a live graph showcasing the historical correlation between two stocks.
- Implemented upper and lower-bound indicators based on correlation analysis, enabling real-time monitoring of key financial metrics.

EDUCATION

B.M.S. College of Engineering Bangalore, India

Bachelor of Engineering – Electronics & Instrumentation Engineering; GPA: 8.3/10 2020 – 2024

S.Tech IT School Rajasthan, India

Higher Secondary School Certificate – PCM; Percentage: 82.6% 2019 – 2020

TECHNICAL SKILLS

Languages & Frameworks: Python, SQL, PyTorch, TensorFlow, Scikit-Learn, Keras, Pandas, NumPy, OpenCV

Generative AI & LLMs: LangChain, LangGraph, Google ADK, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), Prompt Engineering, Embeddings, AI Agents, Multi-Agent Systems, Fine-Tuning, Retrieval Pipelines, Hybrid Search, Reranking

AI & Machine Learning: Deep Learning, Generative AI, Large Language Models (LLMs), Agentic AI, NLP, Computer Vision, Transformers, XGBoost, LightGBM, Hyperparameter Optimization, Document AI (OCR, Extraction), Hugging Face

Data Engineering & Databases: Snowflake, Kafka, BigQuery, PostgreSQL, MongoDB, MySQL, Redis, Elasticsearch, Vector Databases, Data Pipelines, Exploratory Data Analysis (EDA), Time-Series Forecasting

Cloud, Infrastructure & Deployment: Git, GitHub, GCP (Vertex AI), AWS, Azure ML, Docker, Kubernetes, CI/CD, GitHub Actions, Linux, Shell Scripting, FastAPI, Flask, REST APIs, Streamlit

CERTIFICATIONS AND ACHIEVEMENTS

Associate Cloud Engineer – Google Cloud (2024): Successfully completed the Google Cloud Associate Cloud Engineer certification, validating expertise in designing, deploying, and managing cloud solutions using GCP services.